Moller scattering 2

Here we solve path integrals to find $\mathcal{M}_1$ and $\mathcal{M}_2$ that were used in the previous section.

$\mathcal{M}_1$ is the amplitude for $1 \to 3$, $2 \to 4$.

$$\mathcal{M}_1 = i \iint \bar{\psi}_4(y)(-ie\gamma^\mu)\psi_2(y)G(x,y)\bar{\psi}_3(x)(-ie\gamma_\mu)\psi_1(x)\, d^4x\, d^4y$$

Function $G(x,y)$ is the photon propagator

$$G(x,y) = \frac{i}{(2\pi)^4} \int \frac{\exp(-ip\cdot(x-y))}{p^2 + i\epsilon}\, d^4p$$

The normalized wave functions are (V is volume)

$$\begin{aligned}
\psi_1(x) &= \sqrt{2EV}u_1 \exp(-ip_1\cdot x) \\
\psi_2(x) &= \sqrt{2EV}u_2 \exp(-ip_2\cdot x) \\
\psi_3(x) &= \sqrt{2EV}u_3 \exp(-ip_3\cdot x) \\
\psi_4(x) &= \sqrt{2EV}u_4 \exp(-ip_4\cdot x)
\end{aligned}$$

Substitute for the wave functions in $\mathcal{M}_1$ and factor the spinor products.

$$\mathcal{M}_1 = -\frac{e^2}{4E^2V^2}(\bar{u}_4\gamma^\mu u_2)(\bar{u}_3\gamma_\mu u_1) \\ i \iint \exp(i(p_4-p_2)\cdot y)\, G(x,y) \exp(i(p_3-p_1)\cdot x)\, d^4x\, d^4y$$

Substitute for $G(x,y)$.

$$\mathcal{M}_1 = \frac{e^2}{4E^2V^2}(\bar{u}_4\gamma^\mu u_2)(\bar{u}_3\gamma_\mu u_1)\frac{1}{(2\pi)^4} \\ \iiint \exp(i(p_4-p_2)\cdot y)\, \frac{\exp(-ip\cdot(x-y))}{p^2+i\epsilon} \exp(i(p_3-p_1)\cdot x)\, d^4x\, d^4y\, d^4p$$

Combine exponentials.

$$\mathcal{M}_1 = \frac{e^2}{4E^2V^2}(\bar{u}_4\gamma^\mu u_2)(\bar{u}_3\gamma_\mu u_1)\frac{1}{(2\pi)^4} \\ \iiint \frac{\exp(i(p_4-p_2+p)\cdot y)\exp(i(p_3-p_1-p)\cdot x)}{p^2+i\epsilon}\, d^4x\, d^4y\, d^4p$$

Solve the integral over x.

$$\int \exp(i(p_3-p_1-p)\cdot x)\, d^4x = (2\pi)^4\delta^4(p_3-p_1-p)$$

Solve the integral over y.

$$\int \exp(i(p_4 - p_2 + p) \cdot y)\, d^4y = (2\pi)^4 \delta^4(p_4 - p_2 + p)$$

Hence

$$\mathcal{M}_1 = \frac{e^2}{4E^2V^2}(\bar{u}_4\gamma^\mu u_2)(\bar{u}_3\gamma_\mu u_1)(2\pi)^4 \int \frac{\delta^4(p_4 - p_2 + p)\delta^4(p_3 - p_1 - p)}{p^2 + i\epsilon}\, d^4p$$

The integrand vanishes except for

$$p = -p_4 + p_2 = p_3 - p_1$$

Hence

$$\mathcal{M}_1 = \frac{e^2}{4E^2V^2}(\bar{u}_4\gamma^\mu u_2)(\bar{u}_3\gamma_\mu u_1)(2\pi)^4 \frac{\delta^4(p_4 + p_3 - p_2 - p_1)}{(p_3 - p_1)^2}$$

$\mathcal{M}_2$ is the amplitude for $1 \to 4$, $2 \to 3$.

$$\mathcal{M}_2 = i \iint \bar{\psi}_3(y)(-ie\gamma^\mu)\psi_2(y)G(x,y)\bar{\psi}_4(x)(-ie\gamma_\mu)\psi_1(x)\, d^4x\, d^4y$$

It follows that

$$\mathcal{M}_2 = \frac{e^2}{4E^2V^2}(\bar{u}_3\gamma^\mu u_2)(\bar{u}_4\gamma_\mu u_1)(2\pi)^4 \int \frac{\delta^4(p_3 - p_2 + p)\delta^4(p_4 - p_1 - p)}{p^2 + i\epsilon}\, d^4p$$

The integrand vanishes except for

$$p = -p_3 + p_2 = p_4 - p_1$$

Hence

$$\mathcal{M}_2 = \frac{e^2}{4E^2V^2}(\bar{u}_3\gamma^\mu u_2)(\bar{u}_4\gamma_\mu u_1)(2\pi)^4 \frac{\delta^4(p_4 + p_3 - p_2 - p_1)}{(p_4 - p_1)^2}$$

For the momentum vectors given in the previous section we have

$$\delta^4(p_4 + p_3 - p_2 - p_1) = \delta^4(0) = 1$$

Hence

$$\mathcal{M}_1 = \frac{(2\pi)^4}{4E^2V^2}\frac{e^2}{(p_3 - p_1)^2}(\bar{u}_4\gamma^\mu u_2)(\bar{u}_3\gamma_\mu u_1)$$

$$\mathcal{M}_2 = \frac{(2\pi)^4}{4E^2V^2}\frac{e^2}{(p_4 - p_1)^2}(\bar{u}_3\gamma^\mu u_2)(\bar{u}_4\gamma_\mu u_1)$$

In practice the leading coefficient is absorbed in the calculation of $d\sigma$ hence we use the formulas

$$\mathcal{M}_1 = \frac{e^2}{(p_3 - p_1)^2}(\bar{u}_4\gamma^\mu u_2)(\bar{u}_3\gamma_\mu u_1)$$

$$\mathcal{M}_2 = \frac{e^2}{(p_4 - p_1)^2}(\bar{u}_3\gamma^\mu u_2)(\bar{u}_4\gamma_\mu u_1)$$

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